

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed systems at scale across consumer products and infrastructure. Currently staff engineer at **YouTube** managing video delivery and streaming pipelines serving billions of daily requests. Previously led platform teams at **Whatnot** scaling marketplace infrastructure and managed teams building search and recommendation systems at **Google**. Demonstrated impact includes designing systems processing millions of queries per second and leading teams of 8 engineers across time zones. Seeking next challenge in high-scale distributed systems or platform engineering with focus on reliability and performance. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, Java

Distributed Systems: Spanner, Bigtable, BigQuery, Pub/Sub, Kafka, Microservices

Infrastructure: Kubernetes, Docker, Terraform, GCP, Linux, C++

Data Engineering: ETL Pipelines, Dataflow, Apache Beam, SQL, Python

Platform Engineering: API Design, gRPC, REST, Caching, Load Balancing, On-call

Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Owned video delivery infrastructure processing billions of daily requests, migrating legacy pipelines to **Kubernetes**-based microservices on **GCP** and reducing operational overhead by 40%.
- Designed **Pub/Sub**-driven event processing system handling 500K messages per second for real-time analytics and viewer engagement metrics.
- Led cross-functional initiative to standardize **gRPC** APIs across 15 video serving teams, improving developer velocity and reducing integration bugs by 60%.
- Established on-call rotation and runbook standards for streaming infrastructure, achieving 99.95% availability across video delivery pipeline.
- Partnered with mobile and web clients to optimize adaptive bitrate algorithms, decreasing buffering events by 35% for live streams.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Architected marketplace platform serving 10M+ users, rebuilding core transaction processing with **Microservices** pattern and **PostgreSQL** to handle 3x traffic growth.
- Built real-time inventory management system using **Redis** and **Kafka**, eliminating oversell issues that previously caused 2% transaction failures.
- Led migration of monolith application to **Kubernetes** clusters, improving deployment frequency from weekly to multiple daily releases.
- Mentored 4 engineers on distributed systems design and **Linux** performance tuning, elevating team-wide code review quality.
- Reduced p99 latency from 800ms to 200ms on listing API by implementing **Caching** layer and database query optimization.

Staff Software Engineer / Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers building search infrastructure serving **BigQuery** metadata and enterprise analytics queries across Google's internal data platforms.
- Designed **Spanner** schema migration strategy supporting 10PB data warehouse, enabling zero-downtime schema changes for tables with billions of rows.

- Built **ETL pipeline** processing 50TB daily for search quality metrics, replacing manual batch jobs with automated **Dataflow** workflows.
- Delivered migration of search backend services from **Bigtable** to **Spanner**, improving query consistency and reducing cross-region replication lag by 70%.
- Established performance budgeting process for search APIs, identifying and eliminating bottlenecks that reduced compute costs by \$2M annually.

Software Engineer | Google | 02/2011 - 02/2015

- Developed indexing infrastructure for Google Maps using **C++** and **MapReduce**, processing geographic data for 100M+ users worldwide.
- Optimized geospatial query execution paths in **Bigtable**, improving point-in-polygon search latency by 40% for map rendering.
- Contributed to open-source **Protocol Buffers** compiler improvements, enhancing code generation for mobile platforms.
- Implemented end-to-end testing framework for map tile generation, reducing production incidents from indexing errors by 80%.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Built core components for **Bing** search index using **C++** and distributed computing frameworks, supporting sub-second query response times.
- Developed ranking algorithm evaluation pipeline enabling A/B testing of 50+ ranking signals in production.
- Optimized inverted index compression algorithms, reducing storage footprint by 25% while maintaining query speed.
- Automated regression testing for search crawler infrastructure, cutting release cycle time from 2 weeks to 3 days.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented internal tooling for test case management using **C#** and **SQL Server**, improving QA team efficiency by 15%.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007